



Research

Python Task

Task 0: Sneaky Log Cleaner (Basic Python)

Imagine you're handed a messy raw log file from a server or an app, and before anyone can analyze it, it needs cleaning up and a bit of personality. Given an input block of log-like text, your function should apply creative transformation rules, such as:

- Masking sensitive data (e.g., replacing all email addresses with [HIDDEN]).
- Normalizing timestamps (e.g., converting 23/08/2025 14:05 into 23rd August 2025, 2:05 PM).
- Adding a fun flag (e.g., every time the word "ERROR" appears, prefix it with).

Write a Python function `transform_logs(input_text: str) -> str` that applies at least 3 transformations of your choice. You can invent your own rules (serious or funny).

Example:

Input: *"User john@mail.com logged in at 23/08/2025 14:05. ERROR: session timeout."*

Output: *"User [HIDDEN] logged in at 23rd August 2025, 2:05 PM. ERROR: session timeout."*

NOTE: There is no single solution to this. All we care about is your creativity and your approach to solving the problem.

Requirements:

Have up-to-date python installed on your system along with the libraries you think are necessary to solve the question.

Task 1: Blog Review (Written Task)

Read any of the given blog/article shared with you and write a short review (200–400 words) on what you understood.

Your review should summarize the key ideas, highlight anything interesting you learned, and reflect on how it connects to your knowledge or interests.

<https://medium.com/@gaurishirke07/prompt-engineering-techniques-streamlining-ai-workflows-for-academic-deadlines-e35219f84240?sharedUserId=gaurishirke07>

<https://medium.com/@samikshapatil486/can-your-camera-tell-if-youre-bored-in-class-bfece6871e58>

<https://medium.com/@devanshik1906/my-exploration-of-rl-concepts-q-learning-dqn-and-a-mountaincar-finale-2da1b3eb5055>

<https://medium.com/@anishchandurkar/building-the-future-of-business-with-agent-ai-2e64ae040da2>

<https://medium.com/@sumbatilinda/deep-learning-part-1-understanding-basic-neural-networks-c9ccdb17a343>

Task 2: Paper Summary (Written Task)

Choose any ONE of the following papers on how to approach reading a research paper:

- [How to Read a Paper — S. Keshav](#)
- [Ten Simple Rules for Reading a Scientific Paper](#)

Write a short summary (200–400 words) explaining the approach the paper describes, what you found most useful, and how you plan to apply it the next time you read a research paper. Be reflective and show curiosity, we want to see your thought process.

Bonus Task: Exploratory Data Analysis (Optional)

Notebook link:

<https://colab.research.google.com/drive/1VBzIsZNPhI-YLWxJGzBfpXvEblB9-RW1?usp=sharing>

Instructions:

1. Open the notebook link.
2. Once it opens in Colab, click File → Save a copy in Drive to make your own editable copy.
3. Run cells one by one by clicking the play button on the left side of each cell.
4. If the notebook uses datasets:
 - a. Download the required CSV file (e.g., from Kaggle) onto your computer first.
 - b. Click the Files icon on the left sidebar, then click the Upload icon and select the CSV file to upload it directly into the Colab session.
 - c. Make sure the file path you use in the code (e.g., `pd.read_csv("filename.csv")`) matches the uploaded file's name exactly.

Note: uploaded files are temporary and will be deleted when the session ends, so re-upload if you reconnect later. If you'd rather not re-upload each time, you can instead upload the CSV to your Google Drive and mount your Drive (Files → Mount Drive).

5. Save progress regularly using File → Save (or it auto-saves to your Drive copy).

References:

EDA: <https://youtu.be/xi0vhXFpegw?si=K7ptKmqS2pRsz3iK>

Requirements:

- Have up-to-date python installed on your system along with the libraries you think are necessary to solve the question.

[This task is optional, however doing this will get you brownie points!]

General Requirements:

Push all the tasks to a public GitHub repository [[YourName_ACMResearchTasks](#)] in the following format.

YourName_ACMResearchTasks/

```
|
|— README.md
|— task0/
|   |— transform_logs.py
|— task1/
|   |— blog_review.md
|— task2/
|   |— paper_summary.md
|— bonus/
|— eda.ipynb
```

[README.md should briefly describe the repository contents and explain how to run Task 0. You can skip the ‘bonus’ folder if you’re not attempting the bonus task.]

(Note: All submissions would be checked for plagiarism, so solve the tasks diligently.)

For any questions related to this task, contact (preferably WhatsApp):

Gauri Shirke: 8169872914

Devanshi Kashyap: 7045683364

Samiksha Patil: 9372493189

Anish Chandurkar: 8104453805